Where the Gains Come From:

A Controlled Study of Feature, Model-Family and Ensemble Interventions on a 700k-row Synthetic Tabular Benchmark

The Time Forever (Kaggle: adamsmith1112)

Playground Series S6E10 — Predicting Airline Satisfaction

October 2026

Abstract

Public leaderboards reward the final score, but practitioners rarely know which interventions actually move it. We study this question on Kaggle Playground S6E10 (699,635 training rows, 21 features, ROC-AUC), where the data are synthesised from a public airline survey, and we measure every intervention on the same five stratified folds with paired DeLong inference and record its public-leaderboard effect. An audit of the data shows why the ceiling is low: 5.66% of rows share their discrete feature signature with another row of the opposite label, 48.3% of the test-time errors are confident errors, and the generator collapsed the delay distribution (mean 1.18 vs. 14.71 minutes), which removes a signal that exists in the original survey. Single-model work saturates at 0.9605 ROC-AUC: capacity sweeps and 16 Optuna trials move the score by less than the paired noise floor of 2.2×10^{-4}. A ladder of more than twenty interventions then separates what transfers from what does not. Cross-fitted target statistics of the exact flight distance are the only large single gain (13.5×10^{-4}); afterwards, *new model families* transfer at a rate above one ($+1.0\times10^{-4}$ cross-validated $\rightarrow +1.5\times10^{-4}$ public), whereas seed averaging ($+0.55 \rightarrow +0.1$), ten folds ($+0.14$), five stacking schemes (±0.3) and five feature-side variants do not. The final eight-family logit blend reaches 0.960901 cross-validated and 0.96062 public, i.e. rank 1,204 of 3,061 teams, and we quantify what the remaining 1.1×10^{-4} to the top decile would cost.

1 Introduction

Kaggle Playground competitions are a convenient laboratory for tabular machine learning: the metric is fixed, the data are public, and thousands of teams optimise the same quantity. They are also a good place to study a question that matters outside competitions: *which kinds of intervention actually improve generalisation, and which only improve the number you are looking at?*

We take the October 2026 episode (S6E10, airline passenger satisfaction, binary ROC-AUC) and run a controlled study. Every intervention is applied to the same five stratified folds, all learned transforms are fitted inside the training part of each fold, and every comparison is reported as a paired difference with a DeLong standard error [2]. The same model is then submitted to the public leaderboard, so each change has two readings: a cross-validated one and a leaderboard one.

Our contribution is empirical rather than algorithmic:

- a data audit that explains the low ceiling of the benchmark (label noise, a generator artefact in the flight-distance support, a collapsed delay distribution, and the semantics of zero-valued ratings);
- a leakage-safe protocol with paired inference and a pre-registered adoption rule, which lets us separate real effects from noise at a resolution of 10^{-4};
- a ladder of more than twenty interventions measured twice (CV and public LB) that yields a simple, quantitative rule: *new model families transfer, everything else mostly does not*;
- a leaderboard analysis that turns the remaining gap into a cost estimate.

2 Related work

What wins on tabular data. Gradient-boosted trees remain the default for medium-sized tabular problems, and careful benchmarks attribute that to their robustness to uninformative features and to rotation invariance rather than to any failure of neural networks [6, 13]. The neural results improved substantially once numerical features were embedded rather than standardised [4], and once architectures were tuned with a modern recipe [3, 7]; parameter efficient ensembling pushes the same family further [5]. Our ensemble is built exactly along that axis: we keep LightGBM as the anchor and look for neural members that are strong *and* decorrelated.

Leakage-safe statistics and paired comparison. Target encoding of a high-cardinality key is classical [10], and its failure mode — target leakage inside the training fold — is well known; we use the inner cross-fitting that makes the estimate honest. Comparing two models on the same rows requires a paired test; the DeLong covariance [2] gives one in closed form, and it is what turns our 10^{-5} differences into statements about noise instead of statements about models. Label noise is usually estimated with confident learning [11]; here the discrete signatures give a direct, model-free estimate that we can check against the residual errors.

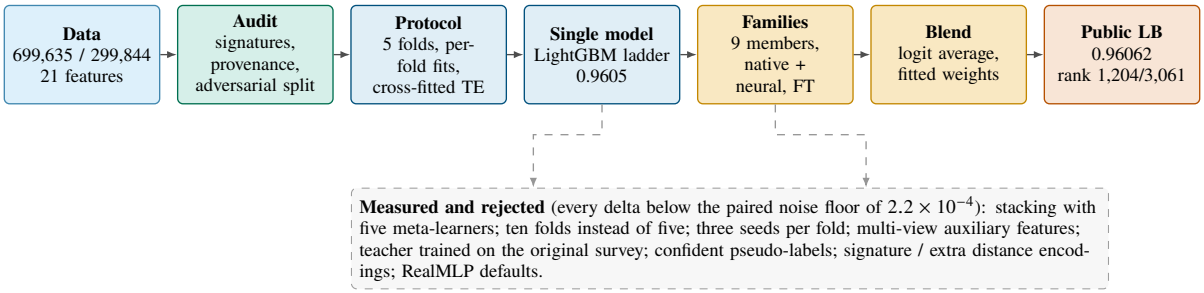


Figure 1: The study at a glance. The upper chain is the adopted pipeline; the dashed box lists the interventions that were measured on the same folds and rejected because their effect stayed inside the noise floor (or, for the leaderboard, inside the tie granularity).

3 Pipeline and experimental design

Figure 1 summarises the study. Data and protocol are fixed first; the single-model ladder is explored until it saturates; model families are then added and blended; every candidate is read twice, on the folds and on the public board, and the rejected ideas are kept in the record because they bound what is worth trying next.

4 Data and audit

4.1 Dataset and split

The data come from the October 2026 Playground episode [8]. The training set has 699,635 rows, the test set 299,844; the identifier is contiguous across the two files (0–999,478) and the positive rate is 44.357%. Twenty-one features describe a passenger (age, flight distance, two delay columns) and a 13-item service questionnaire answered on a 0–5 scale, plus four categorical attributes (gender, customer type, travel type, cabin class). Three facts about the split matter for the rest of the paper. First, an adversarial classifier that tries to separate train from test reaches ROC-AUC = 0.4991, so the two files are exchangeable and cross-validation should track the leaderboard. Second, rating 0 is not the bottom of the scale but a code for *not applicable* (Section 4.4). Third, the data are synthetic (Section 4.3).

4.2 Label noise from discrete signatures

The thirteen ratings and four categorical attributes form a coarse fingerprint: only 437,904 distinct signatures exist for 699,635 rows. Grouping by that signature separates two kinds of rows. About half of the training rows (48.96%) share their signature with at least one other row; within those groups 19.76% of rows disagree with a group member, and flipping the minority members to the majority label would change 5.66% of all labels. Figure 2a breaks this down for coarser groupings: using only the thirteen ratings raises the conflict rate to 30.4%, whereas adding age lowers it to 4.1%, so the continuous columns still carry real resolving information.

The same noise dominates the residual error of a fitted model. For the adopted model, 84.3% of out-of-fold predictions lie below 0.1 or above 0.9, and 4.19% of those confident predictions are wrong; those confident mistakes are 48.3% of all errors. A ranker cannot repair them.

4.3 The generator preserved distances and destroyed delays

The competition data derive from a public survey of 129,880 passengers. Matching the two files on the feature schema gives a precise picture of what the generator kept and what it lost. The support of `Flight Distance` was copied almost verbatim: only 3 of the 3,474 distances in the training set do not occur in the survey, covering 0.0006% of rows. More importantly, the per-distance satisfaction rate survived: across the 2,243 distances with at least 30 synthetic rows, the correlation between the synthetic and the original rate is 0.899 (Spearman 0.896), and the original rate alone scores ROC-AUC = 0.731 on the synthetic data. This is why a target encoding of the exact distance is the single most valuable feature (Section 6): with the default `max_bin=255` the tree model can only approximate 3,474 per-distance rates with 255 bins.

The delay columns were destroyed. In the survey, departure delay has mean 14.71 minutes and 99th percentile 180; in the synthetic data the mean is 1.18 and the 99th percentile 30, with a point-biserial correlation of −0.036 against the label (−0.051 in the survey). Any delay-based feature therefore has nothing to work with, which matches the negative ablation in Figure 5.

4.4 Rating zero is a missing-value code

Figure 2b shows the conditional satisfaction rate for each value of four ratings. The curve is not monotone, and it is not noise: a Wi-Fi rating of 0 is reported by 1.87% of passengers and 88.7% of them are satisfied, while a rating of 1 corresponds to 39.1%. Online boarding has the same shape (62.0% at 0, 10.5% at 1, 87.5% at 5). Two consequences follow. Trees can exploit the pattern with a single split at ≤ 0, so no "is-zero" feature is needed (we tested it: −0.04

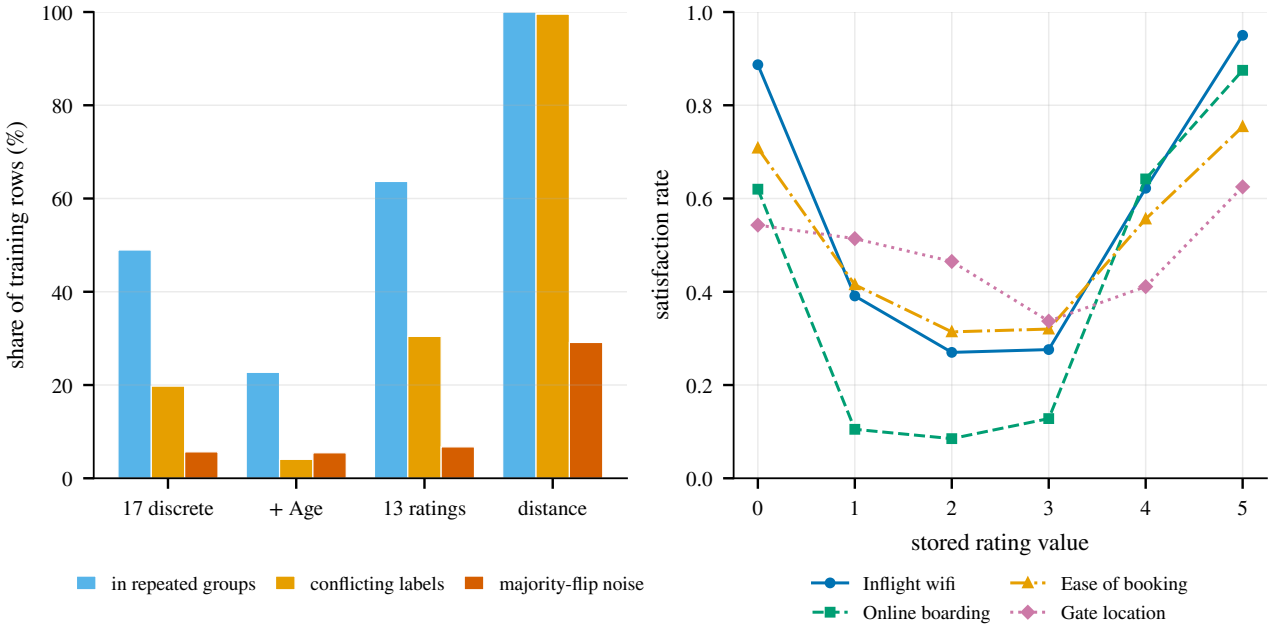


Figure 2: Left: share of training rows whose discrete signature repeats, whose labels conflict inside such groups, and the fraction of labels that a majority vote would flip, for four grouping keys. Right: satisfaction rate as a function of the stored rating. A stored 0 is a "not applicable" code, not a low score: Wi-Fi rating 0 is the *most* satisfied group (88.7%), and every curve is non-monotone.

$\times 10^{-4}$). A linear model, however, must see the ratings one-hot encoded: on the same folds, logistic regression scores 0.9216 with the ratings treated as numbers and 0.9486 with one-hot ratings, a difference of 2.7×10^{-2}.

4.5 No exploitable row-level leak

Because the data are synthetic, one may hope to recover the generating rows of the original survey and inherit their labels. 27.07% of training rows share their full 17-column signature with a survey row, and for those rows the survey label agrees with the synthetic label 93.6% of the time — but the model is better than the source: on the matched rows its ROC-AUC is 0.9428 while the survey label scores 0.9028, and a 50/50 blend of the two drops the overall out-of-fold score from 0.9605 to 0.9592. We also tested for order leaks: consecutive identifiers never share a signature, the autocorrelation of the label is smaller than 10^{-3}, and the mean absolute difference of flight distance between adjacent rows is 1,017 against 1,019 for random pairs. The row order carries no information.

5 Evaluation protocol

5.1 Folds, fits and cross-fitting

All experiments use five stratified folds (seed 42) and, for confirmations, a second five-fold split (seed 2026). Every learned transform — medians for imputation, category vocabularies, target encodings, word frequencies — is fitted on the training part of the fold only. Target statistics use `sklearn`'s `TargetEncoder` with `smooth='auto'` and an inner five-fold cross-fit [10], so a training row never sees its own label through the encoding. Each fold yields out-of-fold probabilities; the headline number is the mean of the five fold scores, and we also report the pooled out-of-fold ROC-AUC.

5.2 Paired inference and the noise floor

Two models scored on the same out-of-fold predictions are highly correlated, so the relevant uncertainty is the standard error of their *difference*. We use the DeLong placement-value covariance [2], which gives a closed form for $\mathrm{SE}[\widehat{\mathrm{AUC}}_a - \widehat{\mathrm{AUC}}_b]$. On this data set the single-model ROC-AUC has a standard error of about 8.5×10^{-4}, but the paired standard error of a difference is only 2.2×10^{-4}–2.7×10^{-4}. That number is the resolution of the whole study: *any change below* 2.2×10^{-4} *cross-validated is indistinguishable from noise on one five-fold split.* For comparison, retraining the same model with a different seed moves the five-fold mean by 0.9×10^{-4} and the public score by 1.2×10^{-4}, which agrees.

5.3 An adoption rule fixed before the experiment

To avoid the usual drift of adopting whatever looks better, the rule is written into the code before a candidate is trained: a candidate is adopted only if it improves the mean fold ROC-AUC on both splits with at least four of five folds ahead, and the paired test on the primary split returns $p < 0.10$. The threshold is deliberately looser than the $p < 0.05$ used for scientific claims, because ensembling is a variance-reduction mechanism with a known sign; but it still blocks the

Table 1: Baselines on the five fixed folds. Mean and population standard deviation of the fold ROC-AUC, pooled out-of-fold ROC-AUC and log loss.

Model	mean fold ROC-AUC	std	OOF ROC-AUC	OOF log loss
Constant prior	0.50000	0.00000	0.50000	0.68677
Logistic regression	0.92156	0.00043	0.92156	0.34424
CatBoost (500 it.)	0.95770	0.00066	0.95770	0.23057
XGBoost (500 it.)	0.95846	0.00061	0.95845	0.22842
LightGBM (500 it.)	0.95862	0.00059	0.95861	0.22789

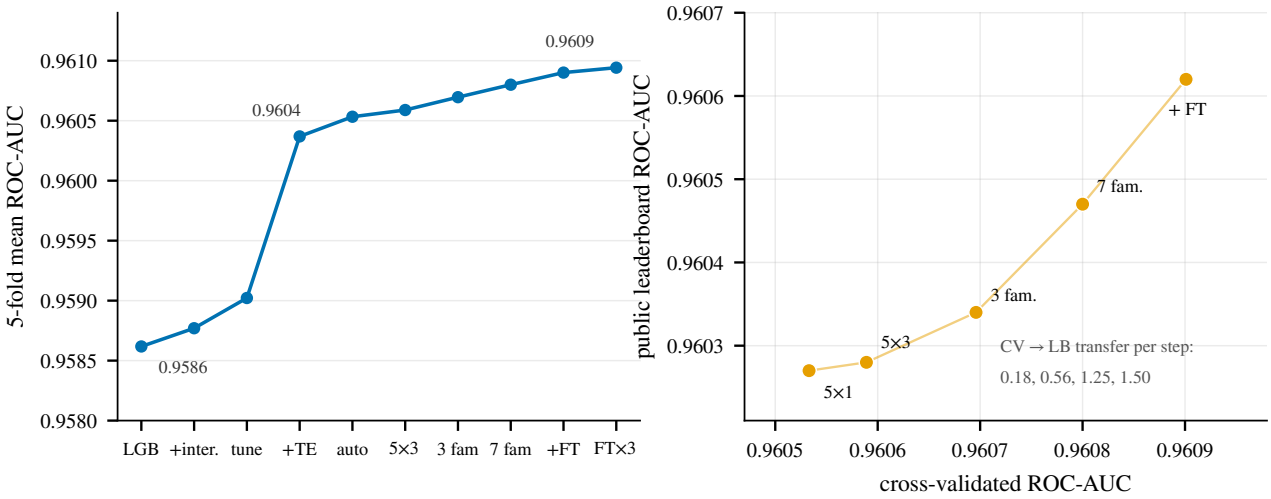


Figure 3: Left: five-fold mean ROC-AUC along the modelling path. Right: the same models on the public leaderboard. Each submitted model sits about 3×10^{-4} below its cross-validated score — the constant offset between the two readings — but the *per-step* transfer grows with the diversity of the ensemble.

10^{-5}-scale changes. When the owner of the repository overrode the rule, the override was recorded in the model-selection file together with the observed p, which keeps the audit trail honest.

6 The single-model ceiling

6.1 Baselines

Table 1 reports the five baselines under the protocol above. The gradient boosting models are close to each other (LightGBM [9] 0.9586, XGBoost [1] 0.9585, CatBoost [12] 0.9577 with its default 500 iterations), the linear model is far behind on numeric ratings (0.9216), and the constant predictor defines the trivial floor.

6.2 What moved the single model, and what did not

The ladder in Figure 3a starts at 0.9586 for a raw LightGBM. Adding five crossed categorical features (travel×cabin, cabin×boarding, . . .) gives 1.6×10^{-4} and 16 Optuna trials 2.4×10^{-4}; both are inside or at the edge of the noise band. The first large step is the cross-fitted target statistics of the flight-distance keys: $+13.5 \times 10^{-4}$ ($p = 5 \times 10^{-4}$), which is the only single change in the whole study that clears the noise floor by a wide margin. Smoothing those encodings with the empirical-Bayes default adds 1.65×10^{-4} ($p = 0.46$); we kept it because it also minimised the tuned objective, but we do not claim it is a real improvement.

Capacity is saturated. Doubling the number of trees (500 → 1,000) costs 0.9×10^{-4}; reducing the leaves from 63 to 31 costs 2.1×10^{-4}; a 127-leaf model at learning rate 0.03 peaks at 500 trees and then degrades; a 255-leaf model at rate 0.02 degrades monotonically (0.9597 at 500 trees down to 0.9585 at 3,000). The validation curve flattens at about 400 trees. The sixteen Optuna trials that produced the adopted configuration are spread over 9.5×10^{-4}, and the top two trials differ by 10^{-5} — a ranking of noise.

7 Model families and blending

Once single-model capacity is exhausted, the only remaining mechanism with a plausible sign is to combine models whose errors are not the same. We therefore trained nine members under the same folds and transforms: LightGBM with the adopted target statistics, four other boosted configurations (XGBoost, CatBoost with 2,500 GPU iterations and native categorical handling, LightGBM with linear leaves, LightGBM with DART dropout), two multilayer perceptrons (one on the raw frame with embeddings, one on the 33 encoded numeric features), an FT-Transformer [3], and — as a

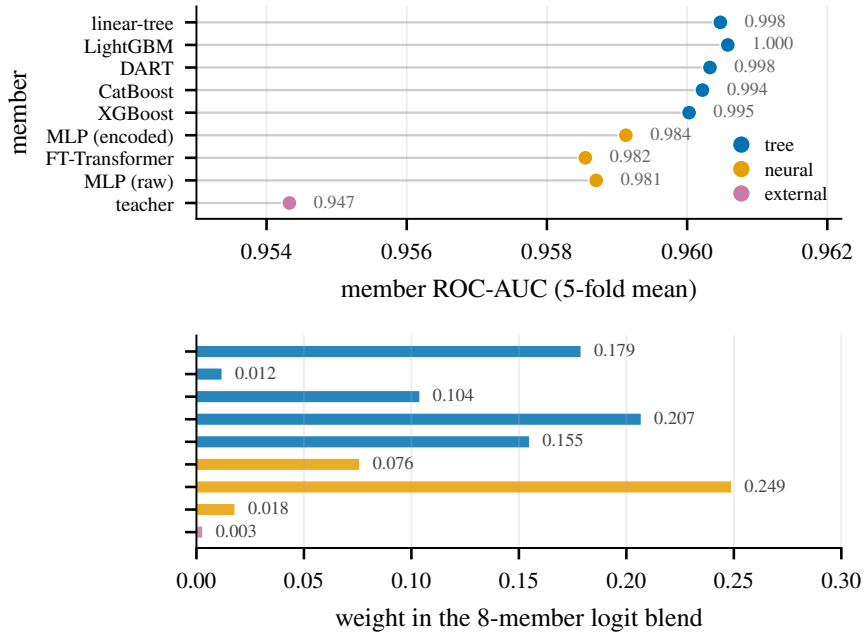


Figure 4: Members of the eight-family blend. Top: five-fold mean ROC-AUC with the residual correlation against LightGBM printed next to each point. Bottom: weight fitted on the logits. Strength and weight are almost independent: the two MLPs and the FT-Transformer absorb the weight that the boosted trees cannot supply.

Table 2: Ensemble ladder. "Public" is the leaderboard score of the submitted file; the transfer column is Δpublic/ΔCV for that step.

Configuration	CV	OOF	public	rank	ΔCV ($\times 10^{-4}$)	transfer
5 folds × 1 seed	0.960533	0.960526	0.96027	1331/2977	—	—
5 folds × 3 seeds	0.960589	0.960580	0.96028	1340/2991	+0.55	0.18
3 families (LGB+XGB+MLP)	0.960696	0.960688	0.96034	1307/3000	+1.07	0.56
7 families	0.960800	0.960793	0.96047	1254/3013	+1.04	1.25
8 families (+FT-Transformer)	0.960901	0.960891	**0.96062**	**1204/3061**	+1.00	1.50
8 families (+FT×3 seeds)	**0.960942**	0.960932	0.96061	—	+0.42	−0.02

deliberately different information source — a model trained only on the 129,880 original survey rows.

The FT-Transformer. Each feature becomes a token: an embedding for each of the 17 discrete fields (including the 3,474-level flight distance), a linear projection for each of the four continuous fields, and a two-level embedding for the arrival-delay missing flag. A learnable class token is prepended and three pre-normalised transformer blocks ($d = 64$, four heads, FFN width 128, dropout 0.1) process the 23 tokens; a linear head on the class token outputs the logit. Training uses AdamW and a one-cycle schedule for 20 epochs at batch 4,096, which costs 97 seconds per fold on a laptop GPU — cheap enough for five folds in eight minutes.

Weights follow decorrelation, not strength. Figure 4 shows the two quantities that matter for a member: its own score and the correlation of its residuals with the LightGBM member. Fitting non-negative weights that sum to one on the logits (Section 8) gives the lower panel. The best single model, LightGBM, ends up with weight 0.012, while CatBoost (0.207), `linear_tree` (0.179), XGBoost (0.155) and the FT-Transformer (0.249) carry the ensemble. The FT-Transformer is the clearest case: it is 2.0×10^{-3} weaker than LightGBM alone and yet takes the largest weight, because it is the most different strong member.

Table 2 traces the ensemble. Three families (LightGBM, XGBoost, MLP) add 1.07×10^{-4} over the multi-seed single model; five more members 1.0×10^{-4}; the FT-Transformer another 1.0×10^{-4}. Each of these steps is at the noise floor on its own, but they are *cumulative* and each is confirmed by the leaderboard, which is the point of Section 9.

8 What does not work

Figure 5 collects the interventions that attribute to a single change, with their measured effect. The lower half of the list is as informative as the upper half.

Table 3: Protocol probe on a fixed 20% split (139,927 rows): the dev part is re-split into k folds, each model is trained on $k-1$ of them and the predictions are averaged before scoring. "Train fraction" is relative to the full training file.

Configuration	train fraction	probe ROC-AUC	Δ ($\times 10^{-4}$)	time
5 folds × 1 seed	0.64	0.961305	—	47 s
5 folds × 3 seeds	0.64	0.961356	+0.55	96 s
10 folds × 1 seed	0.72	0.961319	+0.14	106 s
10 folds × 3 seeds	0.72	0.961355	+0.50	211 s

Table 4: Cross-fitted stacking against the fixed-weight blend (0.960800). "z" is the paired DeLong statistic; every value is statistically indistinguishable from zero.

Meta-learner	fold mean	Δ ($\times 10^{-4}$)	folds ahead	z
Logistic on logits (C = 0.01–10)	0.960797	−0.04	1/5	−0.6
LightGBM on 7 logits	0.960799	−0.01	3/5	−0.3
LightGBM on logits + spread	0.960826	+0.26	4/5	−0.2
LightGBM on logits + raw features	0.960811	+0.11	3/5	−0.2
Per-segment weight vectors (6)	0.960810	+0.09	3/5	+0.1

Variance control. Averaging three seeds per fold buys 0.55×10^{-4} in CV and 0.1×10^{-4} on the leaderboard. Ten folds instead of five buys 0.14×10^{-4} cross-validated: the probe in Table 3 shows that training each model on 72% instead of 64% of the data helps, but the extra averaging over ten models cancels it (10 folds × 3 seeds: 0.961355; 5 folds × 3 seeds: 0.961356 on the same probe split). Strengthening the best member behaves the same way: three seeds lift the FT-Transformer from 0.958548 to 0.959258 and the blend from 0.960901 to 0.960942, yet the public score moves from 0.96062 to 0.96061.

Stacking. We tried five meta-learners on the member logits, all cross-fitted (fit on four folds, predict the held-out one): logistic regression with four values of C, a LightGBM meta-model on the seven logits, the same plus member-disagreement features, the same plus the 21 raw features (conditional stacking), and one weight vector per cabin×travel segment. None of them beats the fixed mean-weight blend (Table 4). The reason is visible in Figure 4: with residual correlations between 0.977 and 1.000 the meta-model has nothing to learn except a global reweighting, and the fitted weights already do that. A meta-model can only pay when the members are both strong and genuinely different.

Feature-side variants. Multi-view auxiliary features (three LightGBMs trained on feature subsets, their cross-fitted logits appended) give 0.57×10^{-4} at $p = 0.81$ and three of five folds: rejected. Pseudo-labelling the 6.6% of test rows the blend is confident about produces a member at 0.960549 with residual correlation 0.9994 — worse than the plain LightGBM it was meant to help. A target encoding of the full 17-column signature, extra distance-segment encodings and `max_bin`= 4095 are all at or below zero.

Weak but different is not enough. The teacher trained only on the original survey is the most decorrelated member we found (residual correlation 0.947, against 0.981 for the best MLP) and it is worthless: at 0.954326 its fitted weight is 0.003. Contribution requires *both* properties — a member needs to be within about 2×10^{-3} of the leader and have a residual correlation below roughly 0.99. Finally, PyTabKit's RealMLP with default hyper-parameters [7] trained for 30 epochs costs 6.9 minutes per fold and reaches only 0.958281, so we did not include it; the published recipe relies on an ensemble of many seeds, which is the family-level move we already exploit through the FT-Transformer.

9 Reading the leaderboard

9.1 The ladder, and why CV under-predicts the gain

Table 2 and Figure 3b tell one coherent story. Every model sits about 3×10^{-4} below its cross-validated score — the offset is remarkably stable (-2.6, -3.1, -3.6×10^{-4} across versions), which is what an exchangeable split predicts. What changes is the *slope*: the same $+1.0 \times 10^{-4}$ of cross-validated improvement is worth $+0.1 \times 10^{-4}$ when it comes from seed averaging, +0.6 when it comes from a three-family blend, +1.3 when it comes from seven families and +1.5 when the FT-Transformer joins. Averaging different *functions* generalises better than averaging one function many times, and the public split (299,844 rows) sees that more clearly than five validation folds of 140,000 rows each.

The counter-example is equally instructive: three-seeded FT-Transformer is the best cross-validated model we built (0.960942) but scored 0.96061, one 10^{-5} below its single-seeded sibling. We therefore kept the leaderboard-best artifact

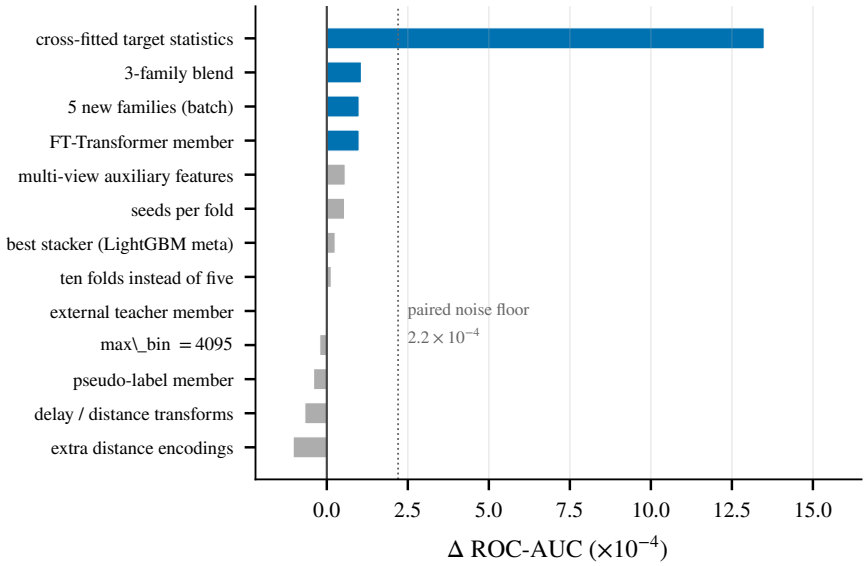


Figure 5: Every intervention measured against the same reference, in units of $\times 10^{-4}$. Blue: adopted. Grey: measured and rejected. The dotted line is the paired noise floor (2.2×10^{-4}); everything to its left is indistinguishable from retraining with another seed.

and recorded both numbers in the model-selection file. Reporting the CV-best number while shipping the LB-best file would have been the easy way to make this section look better than it is.

9.2 What the leaderboard's shape implies

Figure 6 converts the remaining gap into effort: near our band roughly 50 places cost 10^{-4} of ROC-AUC, so the 1.1 $\times 10^{-4}$ that separate us from the top decile (0.96192) is about 550 places. Two further observations from the snapshot of 3,061 teams help calibrate what such a jump would require.

First, the board is dominated by ties. Of 3,013 teams in an earlier snapshot only 749 distinct scores exist, and 130 teams share the single value 0.96213 (ranks 21–150). A 130-way tie at five decimals is not 130 independent discoveries; it is the signature of a shared artifact — published out-of-fold predictions and frozen stacks, which are explicitly distributed in this competition. Second, submission counts increase with rank: the median top-10 team has made 47 submissions against 4 for the field, and the top two have made 178 and 150. Because the public board records a team's best public submission, part of the visible gap is selection, not skill; the private split will settle it.

Finally, the two leaders are outliers in a way that stacking arithmetic does not explain: $+0.64 \times 10^{-4}$ above rank 2 and $+1.04 \times 10^{-4}$ above rank 3, whereas ranks 3–10 span only 0.32×10^{-4}. We could not identify the mechanism from public material, and we ruled out the leak candidates we could test (Section 4.5).

10 Discussion and limitations

A rule of thumb. Across the interventions we measured, the outcome is summarised by one sentence: *changes that alter the hypothesis class transfer to unseen data; changes that reduce the variance of a fixed hypothesis class mostly do not.* The target encoding of the distance is the extreme case of the first kind ($+13.5 \times 10^{-4}$, transfer ≈ 1), a new architecture family is a milder version of it ($+1.0 \times 10^{-4}$, transfer 1.5), and seed averaging, fold count, stackers and pseudo-labels belong to the second kind. The practical consequence is a budget rule: if a change costs more than an hour and is expected to buy less than 3×10^{-4} in CV, it will not be visible on a board whose granularity is 10^{-4}.

The remaining gap is a budget, not a trick. The top decile needs 1.1×10^{-4} more. At 2–5 $\times 10^{-4}$ per qualifying member (strong and residual-correlated below 0.99), that is 20–50 additional members — consistent with the hundreds of members that public stacks combine. We did not pursue the shared-OOF route, which is the only way to reach that scale within a competition, because it cannot be reproduced from a repository that does not own the other teams' predictions.

Limitations. (i) One competition, one metric, one month; the transfer rates are specific to a 300k-row public split and to a synthetic generator whose noise is unusually high. (ii) The public board is a moving target: team counts changed from 2,977 to 3,061 during the study, and ties are resolved by submission time, so rank changes smaller than about 20 places are not meaningful. (iii) Part of the top of the board is selection over many submissions and cannot be attributed to modelling. (iv) Our adoption gate was overridden twice by the repository owner; both overrides are recorded with the observed p-values rather than hidden. (v) The study uses a single GPU and a single machine; timings are not comparable across hardware.

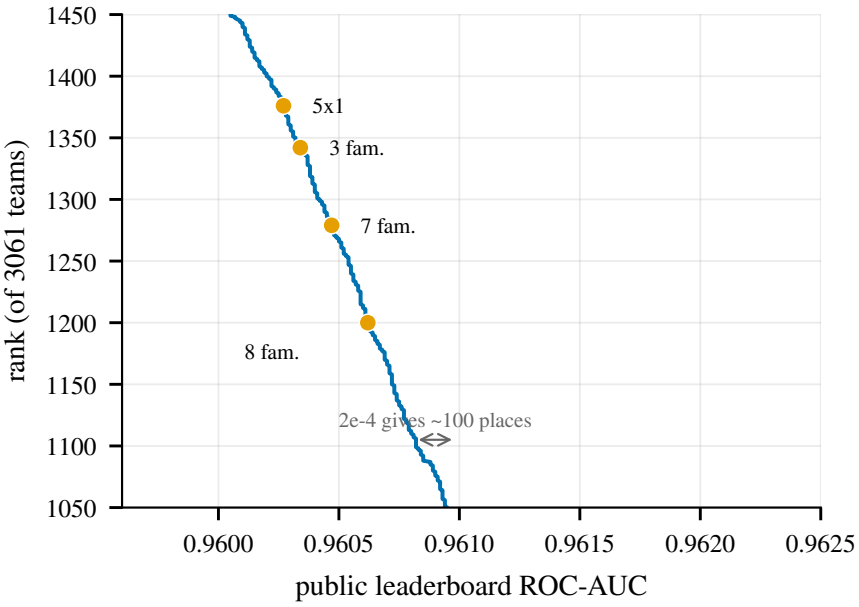


Figure 6: Public leaderboard rank against score, zoomed on the band around our submissions. Near this score, 2×10^{-4} of ROC-AUC buys roughly one hundred places; the top decile is 1.1×10^{-4} away.

11 Reproducibility

Every number in this paper comes from a command in the repository. Folds and their hashes are stored with the preprocessors; model-selection files record weights, seeds, cross-validated scores, confirmation splits and the adoption gate.

Stage	Command
Preprocess and fix folds	`python main.py preprocess ; main.py cv`
Baselines and ablations	`main.py baseline ; main.py features`
Target statistics	`main.py targetstats ; main.py explore`
Hyper-parameter search	`main.py tune -trials 16`
Multi-seed final model	`main.py final -seeds 42 1042 2042`
Multi-family blend	`main.py multimodel -weights 0.8 0.1 0.1`
Data audit	`python analysis/audit.py`
Tests	`python -m unittest discover -s tests`

The FT-Transformer member, the stacking study and the multi-family weight search were run as recorded experiment scripts; their outputs (out-of-fold predictions, test predictions, weights, paired statistics) are the inputs of the tables above. The 34 unit tests of the repository cover the leakage guarantees on which the protocol relies: per-fold fitting, unknown categories, probability bounds and OOF alignment.

Acknowledgments

An AI research assistant helped orchestrate the experiments and prepare the manuscript. Every number in the tables above is reproducible from the recorded scripts, and every experiment was retained whether or not it succeeded.

References

[1] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016. doi: 10.1145/2939672.2939785.

[2] Elizabeth R. DeLong, David M. DeLong, and Daniel L. Clarke-Pearson. Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach. *Biometrics*, 44(3):837–845, 1988. doi: 10.2307/2531595.

[3] Yury Gorishniy, Ivan Rubachev, Valentin Khrulkov, and Artem Babenko. Revisiting deep learning models for tabular data. In *Advances in Neural Information Processing Systems*, 2021. arXiv:2106.11959.

[4] Yury Gorishniy, Ivan Rubachev, and Artem Babenko. On embeddings for numerical features in tabular deep learning. In *Advances in Neural Information Processing Systems*, 2022. arXiv:2203.05556.

[5] Yury Gorishniy, Akim Kotelnikov, and Artem Babenko. TabM: Advancing tabular deep learning with parameter-efficient ensembling. In *International Conference on Learning Representations*, 2025. arXiv:2410.24210.

[6] Léo Grinsztajn, Edouard Oyallon, and Gaël Varoquaux. Why do tree-based models still outperform deep learning on tabular data? *arXiv preprint arXiv:2207.08815*, 2022.

[7] David Holzmüller, Léo Grinsztajn, and Ingo Steinwart. Better by default: Strong pre-tuned MLPs and boosted trees on tabular data. In *Advances in Neural Information Processing Systems*, 2024. arXiv:2407.04491.

[8] Kaggle. Playground series S6E10: Predicting airline satisfaction. https://www.kaggle.com/competitions/playground-series-s6e10, 2026. Competition, October 2026.

[9] LightGBM developers. LightGBM documentation: Parameters. https://lightgbm.readthedocs.io/en/latest/Parameters.html, 2026. Accessed 2026-10-25.

[10] Daniele Micci-Barreca. A preprocessing scheme for high-cardinality categorical attributes in classification and prediction problems. *ACM SIGKDD Explorations Newsletter*, 3(1):27–32, 2001. doi: 10.1145/507533.507538.

[11] Curtis G. Northcutt, Lu Jiang, and Isaac L. Chuang. Confident learning: Estimating uncertainty in dataset labels. *Journal of Artificial Intelligence Research*, 2021. arXiv:1911.00068.

[12] Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. CatBoost: unbiased boosting with categorical features. *arXiv preprint arXiv:1706.09516*, 2018.

[13] Ravid Shwartz-Ziv and Amitai Armon. Tabular data: Deep learning is not all you need. *Information Fusion*, 81: 84–90, 2022. doi: 10.1016/j.inffus.2021.11.011.